Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering - M.E in Computer Science and Engineering (Artificial Intelligence and Machine Learning), programare as follows

PEO No.	Education Objective
PEO 1	Acquire solid mathematical and computational skills essential for understanding, applying, and developing modern artificial intelligence and machine learning algorithms.
PEO 2	Produce industry-ready graduates with practical experience in structuring machine learning projects using state-of-the-art software.
PEO 3	Address multi-disciplinary challenges through coursework and projects by adapting to the rapidly advancing developments in artificial intelligence and machine learning approaches.

Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering - M.E in Computer Science and Engineering (Artificial Intelligence and Machine Learning), graduates will be able to:

PO 1	An ability to independently carry out research/investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	Demonstrate a degree of mastery over the area as per the specialization of the program which should be at a level higher than the requirements in an appropriate bachelor program.
PO 4	Identify appropriate and efficient algorithmic approaches for solving real-life challenges using artificial intelligence and machine learning principles and state-of-the art software prevalent in industry and academia.
PO 5	Develop teamwork and leadership skills for addressing challenges of social importance for sustainable societal development using ethical artificial intelligence and machine learning approaches.

1. Course Plan

1.1 Primary Information

Course Name	:	Algorithms and Data Structures for Big Data Lab [BDE 5151]
L-T-P-C	:	0-0-3-1
Contact Hours	:	36 Hours
Pre-requisite	:	Programming with Python or C
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No. of Contact Hours	Program Outcomes (POs)	BL
CO1	Implementation of linked lists, stack and queues	15	PO3	5
CO2	Implementation of binary search tree, sorting and searching, dictionary and Hash Table.	12	PO4	5
CO3	Apply graphs and shortest path techniques for the practical problems.	9	PO4	5

1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Linked List : Implementing Single Linked List	CO1
L2	Linked List : Implementing Double Linked List	CO1
L3	Linked List : Application development using linked lists	CO1
L4	Stack : Implementation and applications of Stack	CO1
L5	Queue: Implementations of Queue	CO1
L6	Tree - Implementation and applications of Tree	CO2
L7	Applications using different search and sorting techniques	CO2
L8	Applications using different search and sorting techniques	CO2
L9	Hash Table - Applications	CO2
LT	Lab Test	CO1, CO2, CO3
L10	Graph representation using list and matrix method	CO3

L11	Graph traversal	CO3
L12	Graph: Shortest path technique	CO3

1.5 References

1. Introduction to Algorithms - Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest. MIT Press.
2. Data Structures and Algorithms - Aho, Hopcroft and Ulmann. Pearson Publishers.
3. Data Structures and Algorithms in Python - Michael T. Goodrich, Roberto Tamassia, and Michael H. Goldwasser. John Wiley & Sons.
4. Data Streams: Algorithms and Applications - S. Muthukrishnan. Foundations and Trends in Theoretical Computer Science archive, Volume 1 Issue 2, August 2005, Pages 117 – 236.

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.